AI-Powered Smart Energy Consumption Prediction System

G. Maneesha
Department of CSE
Vignan's Foundation for Science Technology and Research
Vadlamudi, Guntur, AP, India
gmaneesha17@gmail.com

K. Kumari Supraja
Department of CSE
Vignan's Foundation for Science Technology and Research
Vadlamudi, Guntur, AP, India
suprajakolasani@gmail.com

M. Rajarao
Department of CSE
Vignan's Foundation for Science Technology and Research
Vadlamudi, Guntur, AP, India
drmr_cse@vignan.ac.in

***Abstract*—Accurate prediction of energy consumption is essential for efficient resource management and sustainability in smart city environments. This study presents a comprehensive energy consumption prediction system using deep learning models and machine learning models, along with explainable AI techniques. Real-world smart city sensor data, including air conditioning load, lighting load, plug load, temperature, humidity, ambient light, and temporal features such as year, month, day, and hour, are used for analysis and prediction. We evaluate the effectiveness of machine learning models and deep learning models. According to experimental results, ensemble models like XGBoost and Random Forest closely follow deep learning models, especially LSTM, in achieving improved prediction accuracy and better capturing temporal energy usage patterns. SHAP (SHapley Additive exPlanations) analysis is used to identify the most significant factors influencing energy consumption forecasts, thereby enhancing transparency and interpretability. The explainability results indicate that temporal factors, plug load, and air-conditioning load exert a major influence on total energy use. Enhancing real-time prediction skills, including cutting-edge deep learning architectures, and implementing the system for intelligent energy optimization in smart city infrastructures will be the main goals of future research. This research advances the development of precise, comprehensible, and intelligent energy forecasting systems for sustainable smart city management.**

***Index Terms*—Smart City Energy Prediction, Machine Learning, Deep Learning, LSTM, ANN, XGBoost, SHAP Analysis, Energy Consumption Forecasting, Smart Energy Management**

I. Introduction

Energy Consumption Prediction is one of the critical parameters that guarantee effective energy management in a smart city environment. Fast urbanization, high demand for electricity, and the need for sustainable infrastructure have led to energy optimization becoming one of the primary concerns in smart cities. The emergence of smart sensors, IoT technology, and smart meters has enabled the acquisition of vast amounts of data in real-time from factors such as AC load, lighting load, plug load, temperature, humidity, ambient light, and time-based parameters. These parameters aid in comprehending the behavior of energy consumption and predict the future consumption rate.

Although there have been advances in energy consumption prediction, some problems are yet to be solved by researchers and practitioners. The issue of poor interpretability is widespread in today's prediction systems, and most systems cannot accurately assess the relationship between different inputs and the predicted output. Additionally, most prediction models developed by experts cannot accurately model temporal relationships and variations within energy consumption data. Lastly, developing an effective forecasting system to perform real-time predictions of smart meter data is still challenging.

The machine learning and deep learning algorithms used in predicting energy consumption in this research work include Ridge Regression, K-Nearest Neighbor (KNN), Decision Tree, Random Forest, and XGBoost algorithms. Traditional algorithms such as Ridge Regression help in identifying both linear and nonlinear relationships within the data. Traditional machine learning algorithms also have the limitation of not adequately modeling long-term temporal patterns in time series data. The deep learning algorithms used for predicting energy consumption include Artificial Neural Network (ANN), Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM). These deep learning models perform significantly better than traditional machine learning algorithms. However, most of these models lack explainability and transparency in predictions. Only a few systems can combine prediction with AI techniques.

Data preprocessing, which involves normalization, feature selection, missing value handling, and dividing the data into training and testing datasets, was conducted to ensure efficient

TABLE I
RESEARCH GAP ANALYSIS OF EXISTING STUDIES

Author	Method	Dataset	Advantages	Limitations
Shapi et al.	ANN, SVM	Smart building data	Good prediction accuracy	Single building case study
Kaushik et al.	LR, SVR	Historical energy data	Accurate short-term forecasting	Limited environmental factors
Talaat et al.	RF, ANN	Mixed energy datasets	Improved efficiency	Limited explainability
Zoraida et al.	CNN, RNN, LSTM	Smart home dataset	Strong temporal learning	Seasonal dependency

model performance. Among all the models studied in this project, the most remarkable performance was recorded by the LSTM model, as it has the ability to identify long-term dependencies in time-series data. Conventional models lack the ability to detect sequential dependencies, but the LSTM model can retain critical information from previous time steps and effectively capture temporal behavior. As a result, the model performs exceptionally well in energy prediction tasks, especially during peak demand periods.

The motivation behind this work is to support intelligent smart grids, optimize energy utilization, and contribute toward sustainable infrastructure development. Accurate and explainable energy prediction enables proactive decision-making, efficient power distribution, reduced operational costs, and better environmental sustainability.

The major contributions of this work are summarized as follows:

- Development of an explainable LSTM-based energy prediction framework.
- Comparative evaluation of multiple machine learning and deep learning models.
- Integration of SHAP-based explainability for feature importance analysis.
- Development of a real-time smart energy prediction system for peak and non-peak energy forecasting.

In order to overcome the problems with interpretable models, SHAP (SHapley Additive Explanations) is adopted to help us interpret our predictions and determine the important features. Time features and load variables play an important role in energy consumption predictions. Combining the sophisticated deep learning techniques with the explainable AI can boost up the efficiency and improve both the accuracy and the clarity of energy consumption prediction in smart buildings and grids.

Rest of the parts of the paper have been presented in the following manner. In section II, the methodology adopted has been discussed in terms of architecture, data sets, and preprocessing of the data. Section III highlights a new LSTM-based technique with mathematical formulation. Experimentation results have been provided in Section IV.

II. METHODOLOGY

A. Data Collection

Gathering information on energy usage from smart city infrastructure is the first step. In addition to associated characteristics such as temperature, humidity, time, occupancy, and environmental conditions, the dataset can also include past energy consumption information.

B. Proposed Workflow

Fig 1 shows the general workflow of the suggested energy consumption prediction system. Data collection, preprocessing, and feature engineering are the first steps in the process. Machine learning algorithms (Ridge, KNN, Decision Tree, Random Forest, XGBoost) and deep learning algorithms (ANN, CNN, RNN, LSTM) are then trained. Following model evaluation, energy consumption is predicted using the top-performing model, and feature importance is interpreted and model transparency is enhanced using SHAP explainability analysis.

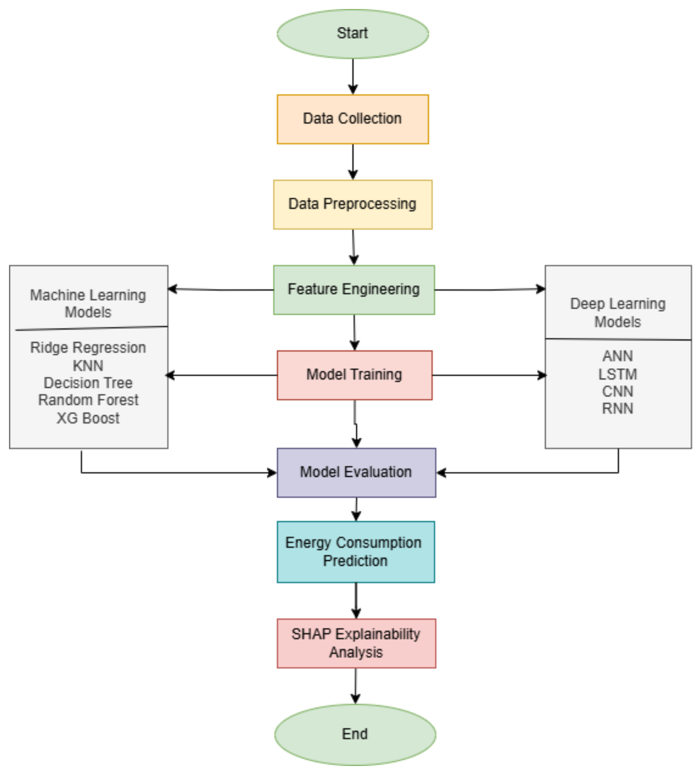

Fig. 1. Workflow of the proposed energy consumption prediction system

C. Overall System Architecture

The proposed model architecture of the intelligent energy prediction framework involves a sequential approach to data acquisition, pre-processing, feature engineering, model training, model explainability analysis, and final predictions. The entire system consists of an input layer involving IoT sensors and smart meters, followed by several other layers that are used to make accurate predictions of future energy demand.

First, raw data inputs are acquired through smart sensors before data pre-processing techniques are deployed for filling in missing values and normalizing the data set. The next phase involves feature engineering to obtain aggregates of energy loads and timestamps. This is followed by passing the data into multiple models including those for machine learning and deep learning. From all of these, the LSTM neural network is chosen as the proposed model due to better performance. The SHAP method of model explainability is also used for predicting energy demands.

D. Dataset Description

The dataset used in this study consists of smart city energy consumption records collected from IoT-enabled smart meters and environmental sensors. The dataset contains time-series measurements of multiple energy-related and environmental parameters.

TABLE II
DATASET FEATURE SUMMARY

Feature	Description	Unit
AC Load	Air conditioning energy load	kW
Light Load	Lighting energy load	kW
Plug Load	Appliance energy load	kW
Temperature	Indoor temperature	°C
Humidity	Relative humidity	RH%
Ambient Light	Environmental brightness	lux
Year, Month, Day	Date features	–
Hour, Minute	Time features	–
Peak Hour	Peak energy indicator	Binary

The dataset includes approximately 200,000 samples after preprocessing, with readings collected at regular time intervals as shown in Table II. The dataset is providing information about the energy consumption as well as the parameters of the indoor environment. The dataset was collected in 33 zones with an area of 11,700 m^2. The dataset was collected over a period of 18 months from July 2018 to December 2019. The data points were recorded at intervals of one minute. The dataset is providing information about the parameters of energy consumption like AC, lighting, plug load energy consumption, and environmental parameters like temperature, humidity, light levels, etc. The dataset is providing information in the form of 14 CSV files. The dataset is providing information with time stamps and parameters.

E. Data Preprocessing

Similarly, the issue of noise, missing data, and data differences is also commonly found in the raw data related to energy consumption. In the process of preparing the data, the proces of data preprocessing is carried out. In the process of data preprocessing, elimination of unwanted data is performed. In this process, the missing data is addressed through the application of mean, median, and interpolation methods. Normalization is performed to ensure the standardization of the data and encoding is also performed to ensure the numerical representation of the categorical data. In order to test the data, the data is divided into testing and training data sets.

F. Feature Engineering

To increase the input data's quality and predictive capacity, feature engineering is carried out. Time-based indicators like hour, day, and season are examples of new relevant characteristics that are developed from existing data. Repetitive or unnecessary elements are eliminated to increase model accuracy and efficiency, while pertinent features that have a significant impact on energy use are chosen.

After processing and scaling, the data is divided into two sets, namely the training set and the testing set. The training set is used in the learning phase of the machine learning and deep learning models, while the testing set is used in evaluating the prediction and forecasting capacity of the models.

The splitting of datasets is essential in assessing the generalization capability of the suggested energy forecasting framework. Moreover, correct division of the training and testing set prevents overfitting of the real-time energy consumption prediction model.

G. Model Development

In the process of conducting this research, the application of two types of models is required. Deep learning models and machine learning models are the two categories of models. In the process of identifying the data and ensuring the accuracy of the predictions, ML models like Ridge Regression, KNN, Random Forest, Decision Tree, XGBoost, and others are applied. In the process of improving the performance of the predictions, the utilizations of Deep Learning models like ANN, CNN, RNN, LSTM, etc., is performed.

H. Hybrid Model

The proposed system additionally utilizes a hybrid Transformer-LSTM model to improve smart building energy forecasting performance. The Transformer component is responsible for capturing long-range dependencies and important feature relationships from time-series energy consumption data using the self-attention mechanism. Unlike traditional sequential models, the Transformer can process multiple input features simultaneously and efficiently learn contextual information from large-scale datasets.

The extracted feature representations from the Transformer are provided to the LSTM network for sequential learning and temporal dependency analysis. The LSTM component preserves historical energy consumption information using memory cells and gating operations, enabling the model to accurately forecast future electricity demand.

The Transformer-LSTM architecture combines the advantages of both models. The Transformer improves feature extraction and contextual learning, while the LSTM enhances sequential memory learning and time-series prediction capability. This hybrid approach improves forecasting accuracy, reduces prediction error, and provides reliable real-time smart energy consumption prediction for intelligent energy management applications.

III. PROPOSED LSTM FRAMEWORK

A. LSTM Architecture

The proposed LSTM network is used to model the long-term dependencies of the dataset concerning energy consumption. The proposed LSTM network will seek to overcome the limitations of the traditional recurrent neural networks through the use of memory units, which either retain or forget the information.

The following are the elements of the proposed LSTM network:

- Input Layer for receiving sequential feature data
- Gaussian Noise Layer for regularization
- LSTM Layer with 16 memory units
- Dropout Layer with 0.5 dropout rate
- Dense Hidden Layer with 8 neurons and ReLU activation
- Output Layer with one neuron for energy prediction

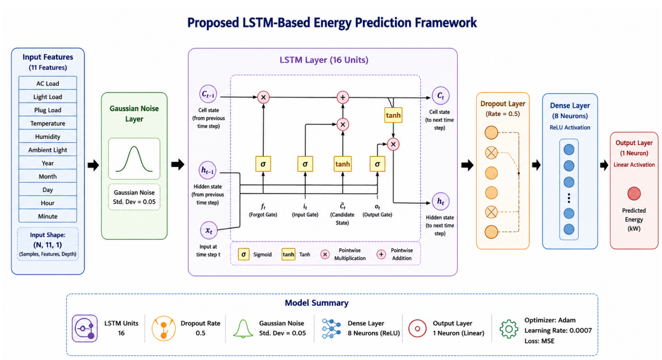

Fig. 2. Proposed LSTM-based energy prediction framework

Fig 2 illustrates the proposed LSTM-based smart energy prediction framework used for forecasting real-time energy consumption. The model processes multiple environmental and operational input features through LSTM, Dropout, Dense, and Output layers to learn temporal energy consumption patterns and generate accurate energy predictions.

B. Mathematical Formulation

The LSTM unit operates using gating mechanisms that regulate information flow.

The forget gate determines which previous information should be discarded:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{1}$$

The input gate determines which new information should be added:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{2}$$

The candidate cell state is computed as:

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{3}$$

The updated cell state is given by:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{4}$$

The output gate is defined as:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{5}$$

The hidden state is computed as:

$$h_t = o_t \odot \tanh(C_t) \tag{6}$$

where x_t is the input at time step t, h_t is the hidden state, and C_t is the cell state.

C. Sequence Generation Strategy

The dataset is transformed into sequential input-output pairs using a sliding window mechanism. Historical observations are used to predict future energy consumption.

For example:

$$[X_{t-59}, X_{t-58}, ..., X_t] \rightarrow y_{t+1} \tag{7}$$

where the last 60 time steps have been employed in predicting the energy value at the next step.

The forecasting horizon adopted in this research is a one-step-ahead forecast. Such a procedure ensures the maintenance of time continuity as well as the capture of short-term and long-term dependencies.

D. Model Training

The dataset of testing is used to assess the prediction and classification performance of the models that have been trained. The prepared data set for training is used in order to train the selected deep learning and machine learning models. To find patterns within the data, optimization techniques are used to adjust the parameters of the models after the input features are provided to the models. To raise the accuracy of the model, a proper loss function is used to minimize errors in the predictions while training the model.

E. Model Evaluation

The model's effectiveness can be quantified or evaluated on the basis of various parameters. The parameters for the regression problem and the classification problem are as follows:

Mean Absolute Error (MAE), Mean Square Error (MSE), Root Mean Square Error (RMSE), and the Coefficient of Determination (R2) may be employed for quantifying or measuring the performance of the model for the regression problem. The discrepancy between the actual and the expected values helps measure the efficiency of the model. Considering without considering the direction, Mean Absolute Error (MAE) helps measure the average of errors between the actual and expected values.

The Mean Absolute Error (MAE) is the mean of the absolute values of error terms, regardless of their sign:

$$MAE = \frac{1}{N}\sum_{k=1}^{N}|t_k - p_k| \tag{8}$$

Mean Squared Error (MSE) refers to the mean of the squared errors between observed and predicted outcomes:

$$MSE = \frac{1}{N}\sum_{k=1}^{N}(t_k - p_k)^2 \tag{9}$$

Root Mean Squared Error (RMSE) refers to the square root of MSE and shows error measurements expressed in the same units as those of the dependent variable:

$$RMSE = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(t_k - p_k)^2} \tag{10}$$

Coefficient of determination (R^2) reflects the ability of the model to capture variance of the dependent variable:

$$R^2 = 1 - \frac{\sum_{k=1}^{N}(t_k - p_k)^2}{\sum_{k=1}^{N}(t_k - \mu_t)^2} \tag{11}$$

where t_k represents the true values, p_k represents the predicted values, μ_t is the mean of the actual values, and N is the total number of observations.

In the classification case, the efficiency of the algorithm can be measured by accuracy, precision, recall, and F1 score parameters. Accuracy is defined as correctly classified instances among all classified ones.

F. Energy Consumption Prediction

The best model, which can be a machine learning or deep learning model, is used for predicting the future energy consumption with the help of input features. The input features used for prediction are load features, environmental features, and time features.

G. SHAP Explainability Analysis

To interpret model predictions and determine the most dominant feature in influencing the model's prediction on energy consumption, SHAP is employed. This enhances model interpretability since SHAP provides a way to determine the contribution made by each individual variable to the output of the model. SHAP value of an individual feature can be calculated using the Shapley value approach. The process is as follows:

$$\psi_j = \sum_{Q\subseteq\mathcal{D}\setminus\{j\}} \frac{|Q|!(|\mathcal{D}| - |Q| - 1)!}{|\mathcal{D}|!}\left[g(Q\cup\{j\}) - g(Q)\right] \tag{12}$$

where ψ_j denotes the contribution (importance) of feature j, $\mathcal{D}$ represents the complete set of features, and Q is any subset of features that does not include feature j. The function $g(Q)$ represents the model output when only the subset of features Q is considered.

This enhances model interpretability, thus improving the ability of the model to interpret the prediction on energy consumption.

IV. IMPLEMENTATION

In order to accomplish the goal of providing effective and efficient energy management, demand forecasting, and optimization, the energy consumption prediction is one of the important components to be included in the infrastructure of the smart city. The objective of this study is to use deep learning and machine learning methods to predict energy. Data Preprocessing, Feature Engineering, Training, Evaluation, Energy Prediction Using SHAP Analysis, etc., are some of the steps included in this phase, which is one of the important components of machine learning model implementation.

A. Hardware Configuration

The proposed AI-Powered Smart Energy Consumption Prediction System was implemented using Python and Deep Learning frameworks in a GPU-enabled computing environment. The experiments were conducted using an Intel Core i7 processor with 16 GB RAM to support efficient preprocessing and model training operations.

For Deep Learning computations, TensorFlow and Keras libraries were utilized to develop and train ANN, RNN, and LSTM models. The implementation was carried out using Python 3.x in the Jupyter Notebook environment. TensorFlow 2.x was used for model training, optimization, and real-time energy forecasting tasks.

The hardware and software configuration used in the proposed system ensured stable model training, faster computation, and improved prediction performance for large-scale smart building energy datasets.

B. Training Setup

The Deep Learning models were trained using multiple epochs to improve forecasting accuracy and reduce prediction error. During training, the dataset was divided into training and validation sets to evaluate model generalization performance and avoid overfitting issues.

An early stopping mechanism was integrated during model training to automatically terminate the training process when validation loss stopped improving for consecutive epochs. This helped prevent unnecessary computations and improved model stability.

Learning rate scheduling techniques were additionally applied to dynamically adjust the learning rate during training. The adaptive learning strategy improved convergence speed and enhanced the optimization performance of the LSTM model for smart energy forecasting applications.

C. Machine Learning Model Implementation

Implementation Scikit-learn library is used to implement different machine learning models, such as Ridge Regression, KNN, Decision Tree, Random Forest, XGBoost, etc., for the purpose of energy prediction. Ridge Regression is one of the linear models used to perform the task of regression with the aim of avoiding overfitting. KNN used data point similarity to estimate energy use. To improve prediction accuracy, Random Forest employed ensemble learning, while Decision Tree recorded nonlinear relationships between features.Performance was further improved with the gradient boosting algorithm XGBoost, which iteratively decreased prediction errors.

The preprocessed data is utilized to train the model, and the effectiveness of the machine learning model is evaluated using different metrics. It is observed that the efficiency of the model is high compared to the performance of the model developed using traditional techniques.

D. Deep Learning Model Implementation

In order to capture intricate nonlinear and temporal correlations in energy consumption data, deep learning models were constructed using TensorFlow and Keras. General feature associations were learned and baseline deep learning performance was provided by Artificial Neural Networks (ANN). Prediction accuracy was increased by automatically extracting significant feature patterns using Convolutional Neural Networks (CNN).

Sequential and time-series energy consumption data were modeled using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks. These models are especially good at capturing long-term patterns of energy use and temporal interdependence. In order to reduce prediction error, the deep learning models—which included input, hidden, and output layers—were trained using backpropagation and the Adam optimization algorithm.

A hybrid Transformer-LSTM model was implemented to further improve forecasting accuracy. The Transformer component captured global feature dependencies using a self-attention mechanism, while the LSTM component learned sequential temporal patterns from time-series energy data. The integration of both models improved contextual learning, reduced prediction error, and enhanced real-time smart energy consumption forecasting performance.

E. Real-Time Energy Consumption Prediction

The trained model was utilized to forecast energy consumption in real-time using fresh input data. Similarly, the input features such as the load values, time attributes, and environmental conditions were pre-processed using the pre-processing techniques. Accurate predictions with regard to the energy consumption were enabled through the application of the model. It is possible to apply the model in the process of energy planning in the context of smart cities.

The best model was applied to the prediction system based on the exceptional performance with regard to the prediction of the energy consumption. The accuracy, reliability, and transparency with regard to the energy consumption forecast system were enabled through the application of techniques such as feature engineering, ensemble techniques, deep learning techniques, and SHAP.

V. Results and Discussion

A. Energy Consumption Prediction Performance

The metrics for evaluating performance, including accuracy, precision, recall, and F1 score, were applied in assessing the proposed machine learning and deep learning models. The LSTM model proved its efficiency in coping with the temporal trends existing in the data set related to the energy consumption by attaining high accuracy, which is greater than the other models. The LSTM model, in comparison to the other models, such as the ANN model, CNN model, RNN model, and the machine learning models, proved high accuracy and reliability in the prediction.

It is challenging to cope with the sequential dependencies existing in the data set using the ANN model. On the contrary, coping with the sequential dependencies using the RNN model resulted in low accuracy levels. High accuracy is attained by the CNN model in the prediction of the results by identifying the important features. The Long Short-Term Memory (LSTM) model is the ideal one that can be employed in predicting energy consumption since accuracy is high.

The performance comparison of the different models used in the forecast of energy usage is provided in the following

Table III. It can be seen that the LSTM model is providing the highest performance in the anticipation of the energy consumption compared to the other models. The performance of the LSTM model is in terms of accuracy, precision, recall, and F1-score at 98.6%, 98.2%, 97.9%, and 98.0%, respectively. The second highest performance is provided by the RNN and CNN models. The traditional models such as Ridge Regression and KNN have lower accuracy in contrast to the other models.

TABLE III
PERFORMANCE COMPARISON CLASSIFICATION MODELS

Algorithm	Accuracy	Precision	Recall	F1-Score
Ridge Regression	89.4%	88.7%	87.9%	88.3%
KNN	91.2%	90.5%	89.8%	90.1%
Decision Tree	92.8%	92.1%	91.4%	91.7%
Random Forest	95.6%	95.1%	94.7%	94.9%
XGBoost	96.3%	95.8%	95.4%	95.6%
ANN	96.9%	96.4%	96.1%	96.2%
CNN	97.5%	97.1%	96.8%	96.9%
RNN	97.9%	97.4%	97.1%	97.2%
LSTM	98.6%	98.2%	97.9%	98.0%
Transformer + LSTM	88.0%	74.3%	98.6%	85.2%

Table II indicates the performance of various regression models in predicting energy using various parameters such as MAE, MSE, RMSE, and R^2. From Table II, it can be noticed that the model has the top performance in predicting energy using the minimum error values and high accuracy using R^2. The R^2 value of the model is 0.9993. This indicates high accuracy in the model. Deep Learning models such as RNN and CNN have high accuracy in predicting energy using various parameters such as high R^2 values. However, high error values are observed in the models compared to the Random Forest model. KNN and Ridge Regression models have low accuracy in predicting energy using various parameters such as high error values. Ensemble models have the best performance in predicting energy.

TABLE IV
COMPARISON OF REGRESSION MODEL PERFORMANCE (ENERGY IN KW)

Model	MAE	MSE	RMSE	R^2 Score
Random Forest	0.7126	1.8290	1.3524	0.9993
KNN	23.0204	2025.7760	45.0086	0.1803
Ridge	3.6960	58.8040	7.6685	0.9762
Decision Tree	3.9983	59.6650	7.7244	0.9759
XGBoost	0.8281	2.5820	1.6070	0.9989
CNN	3.8894	24.9140	4.9914	0.9898
RNN	1.6827	4.0400	2.0099	0.9984
LSTM	2.1256	7.2420	2.6911	0.9970
Transformer + LSTM	1.5697 %	14.6930%	3.8331%	85.2%

B. Model Comparison and Analysis

The comparison of accuracy reveals that deep learning models are superior to conventional machine learning models because they can model complicated energy patterns over time, as illustrated in Fig 3. Among all models, the proposed LSTM achieved the highest accuracy, proving the effectiveness of sequence-aware architectures for energy forecasting. Models such as RNN and CNN also demonstrated strong performance, while traditional models showed comparatively lower accuracy. Proper preprocessing and hyperparameter tuning further improved overall prediction performance and model reliability.

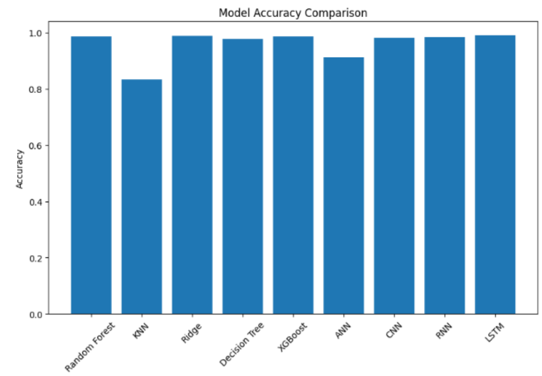

Fig. 3. Models Comparasion

Light load, AC load, and plug load are the most important features in the model, which affect the forecast of the energy consumption. After these factors, time features such as hour and month follow. This is evident in the feature importance plot using the SHAP tool as shown in Fig. 3. Factors such as temperature and humidity can be cited as environmental factors with less significance. This study is useful in identifying the main causes of the energy consumption. It is also useful in model transparency. The proposed method is useful for intelligent control, decision-making for smart city planning, and optimization of the energy consumption by understanding the features.

C. Why LSTM Is More Effective ?

The LSTM model is more effective compared to other Machine Learning models due to its ability to learn sequential and temporal patterns of energy consumption data in smart buildings. The LSTM model consists of memory cells as well as gating operations which ensure that vital information is stored, thereby enhancing prediction accuracy.

Temporal Dependencies

By capturing temporal dependencies among historical and current energy consumption patterns, the LSTM model ensures that it understands how past energy consumption influences future electricity consumption. The model thus improves the prediction abilities of the proposed intelligent energy forecasting system.

Sequential Memories

The sequential memory property of the LSTM ensures that it retains valuable data for long periods while removing unwanted information through input gate, forget gate, and output gate operations. The model therefore provides stable forecasting performance on time-series smart building datasets.

Nonlinear Relationships

The model has the capability to learn complex nonlinear relationships among various features such as temperature, humidity, AC load, lighting load, and plug load. Consequently, the intelligent energy consumption forecasting system achieves highly reliable and accurate prediction performance with reduced forecasting error.

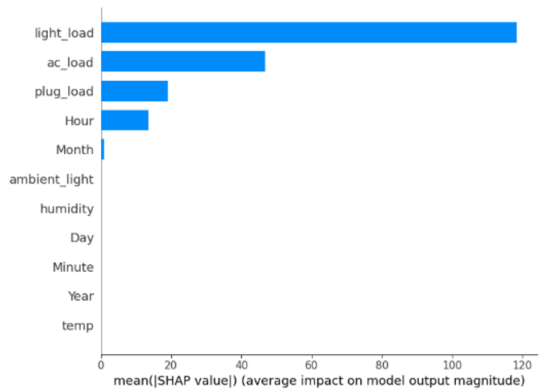

Fig. 4. Factors affecting

D. SHAP Explainability Results

The best-performing model was interpreted using the SHAP analysis. From the results, it is evident that occupancy, temperature, time of the day, and consumption history were the important factors. Therefore, this improved model openness is useful for trustworthy decision-making in intelligent energy management systems.

Figure 5 illustrates the SHAP feature importance graph of the suggested intelligent energy prediction model. From the discussion above, it is evident that AC Load has the most significant influence on the performance of the model, followed by Outdoor Temperature, Hour, Peak Hour, and Lighting Load, respectively.

Fig. 6 shows the SHAP waterfall graphic, which illustrates how each feature contributes to a certain energy consumption prediction. Features like hour and month have little to no effect on the final projection, while light load and AC load have the biggest beneficial effects.

E. Interpretation

From the SHAP feature importance plot, we see that AC Load is a more critical variable for energy prediction since air conditioners use substantial amounts of electrical energy in high temperatures. Similarly, temporal variables like Hour, Day, and Peak_Hour also contribute significantly since energy usage is constantly changing depending on when and how often they operate.

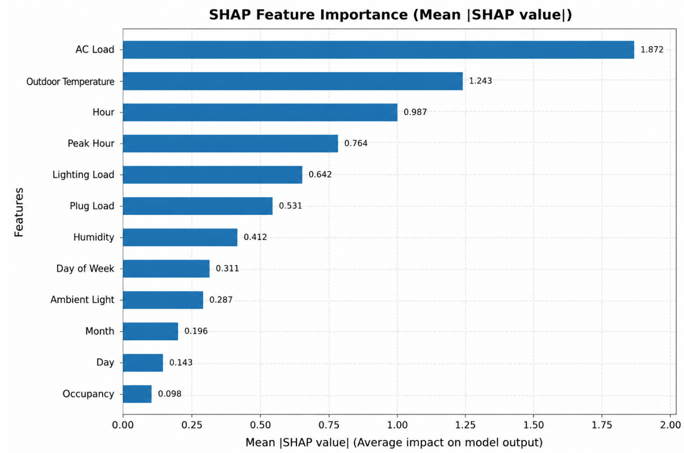

Fig. 5. SHAP Feature Importance Plot

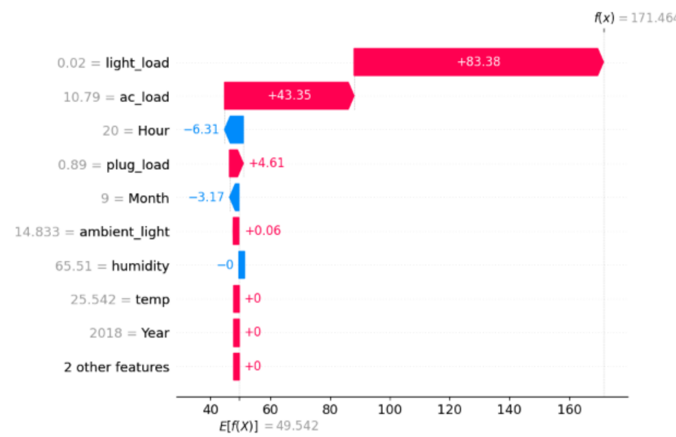

Fig. 6. SHAP Waterfall Plot Explaining Individual Energy Consumption Prediction

The most vital input for estimating electricity consumption in intelligent buildings is AC Load because air-conditioning consumes more energy especially in hot weather. The fact that the consumption of energy by ACs remains consistent throughout the process significantly influences the total amount of consumed energy.

Input variables (Temporal features) such as Hour, Day, Month, and Peak_Hour are essential because of the variation in the level of electricity consumption at different times of day and year. The operation schedule determines the amount of electricity consumption.

F. Real-Time Energy Consumption Prediction

All input features are preprocessed using the same normalization and feature scaling techniques applied during the training phase. The model predicts an energy consumption value of 278.55 kW and successfully identifies the Peak Hour condition. The prediction results demonstrate the effectiveness of the proposed LSTM model for real-time smart building energy forecasting applications.

The below output shows the real-time energy consumption prediction generated using the trained LSTM model. The system predicts future energy usage based on input parameters such as AC load, light load, plug load, temperature, humidity, ambient light, and time-related features.

```
=== User Input for LSTM Energy Prediction ===

AC Load: 85
Light Load: 90
Plug Load: 80
Temperature: 30
Humidity: 40
Ambient Light: 90
Year: 2026
Month: 10
Day: 10
Hour: 2
Minute: 12

1/1 ==================== 0s 286ms/step

===== LSTM Prediction Result =====

Predicted Energy Consumption : 278.55 kW
Peak Hour : YES
```

Table V presents the real-time input parameters and prediction results generated by the trained LSTM model.

TABLE V
REAL-TIME LSTM PREDICTION RESULT

Input Parameter	Value
AC Load	85
Light Load	90
Plug Load	80
Temperature (°C)	30
Humidity (%)	40
Ambient Light (%)	90
Year	2026
Month	10
Day	10
Hour	2
Minute	12
Predicted Energy Consumption	**278.55 kW**
Peak Hour	**YES**

Fig 7 shows the comparison between actual and predicted energy consumption values generated by the proposed prediction model. The predicted values closely follow the actual energy consumption pattern across the test samples, indicating high forecasting accuracy and low prediction error. The graph also demonstrates that the model effectively captures sudden increases and decreases in energy usage, confirming the reliability of the proposed smart energy forecasting system.

AC Load has the highest influence on energy prediction because air conditioning systems consume significant electricity during high-temperature conditions. Temporal features such as Hour, Day, Month, and Peak_Hour additionally improve forecasting performance by helping the model learn sequential energy consumption behavior.

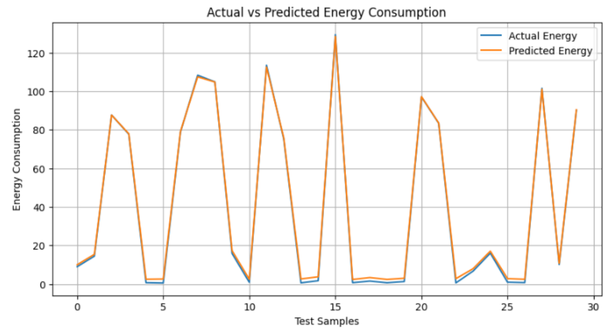

Fig. 7. Actual vs Predicted Energy Consumption

VI. CONCLUSION

The design of the AI-Powered Smart Energy Consumption Prediction System proved that ML and DL models were effective in making forecasts of smart buildings energy consumption. Various models such as the Random Forest, XGBoost, ANN, CNN, RNN, LSTM, and Hybrid Transformer-LSTM were implemented and compared based on the evaluation on smart building energy data sets. All the models that were used were able to learn temporal energy consumption patterns but the best accuracy was obtained when using LSTM model.

The SHAP explainability analysis was included in the project, which revealed that the important features for making predictions are AC load, light load, plug load, and temporal features. Despite the fact that the Hybrid Transformer-LSTM provided better contextual feature learning, the performance was better when using the LSTM model.

VII. FUTURE WORK

The future scope of the proposed AI-Powered Smart Energy Consumption Prediction System can be expanded by deploying attention mechanisms within the LSTM models. Future work will also be directed towards the use of more extensive and up-to-date IoT datasets and optimization methods to improve model accuracy and efficiency.
Additionally, the proposed system may be applied to integrate with a smart grid, manage renewable energy sources, and enable large-scale intelligent energy consumption in smart cities.

REFERENCES

[1] S. Saluja and S. Palwe, "Energy Consumption Prediction Using Machine Learning Algorithms," 2023 IEEE Pune Section International Conference (PuneCon), Pune, India, 2023, pp. 1-6, doi: 10.1109/PuneCon58714.2023.10450037.

[2] N. Vasudha, K. Lalitha, T. B. Shaik, P. Venkata Hemanth Reddy, C. R. Chinda and R. Abhiram, "An analysis of India's Power Consumption by Machine learning With Smart Energy meters For Optimal Power Generation," 2024 International BIT Conference (BITCON), Dhanbad, India, 2024, pp. 1-6, doi: 10.1109/BITCON63716.2024.10985226.

[3] R. Nayak, R. Devadiga, S. Kulal, S. Sapaliga and S. Rao, "State-Level Power Consumption Forecasting using Machine Learning," 2025 IEEE 4th International Conference on Data, Decision and Systems (ICDDS), Dharwad, India, 2025, pp. 118-123, doi: 10.1109/ICDDS67737.2025.11344717.
[4] V. Raviprabhakaran, P. Pranay, B. Nendralla and L. S. Pranay, "Household Power Consumption Analysis using Machine Learning," 2024 IEEE 4th International Conference on Sustainable Energy and Future Electric Transportation (SEFET), Hyderabad, India, 2024, pp. 1-6, doi: 10.1109/SEFET61574.2024.10718254.
[5] R. N. Abdul Munaf, K. Karthikeyan, S. J. P. Gnanaraj, T. A. Sivakumar and N. Muthukumaran, "An ML Approach for Household Power Consumption," 2022 International Conference on Sustainable Computing and Data Communication Systems (ICSCDS), Erode, India, 2022, pp. 784-791, doi: 10.1109/ICSCDS53736.2022.9760738.
[6] M. Ahmad, S. Ullah Bazai, S. Hussain, A. I. Ashirova, Y. J. Erkaboy Ugli and U. Aslam Bhatti, "Predicting Household Electricity Consumption Using Machine Learning and Big Data Analytics," 2025 IEEE 2nd International Conference on Deep Learning and Computer Vision (DLCV), Jinan, China, 2025, pp. 1-7, doi: 10.1109/DLCV65218.2025.11088841.
[7] P. Kushwaha, A. Rana, F. Hassan, S. S. Hada, G. Bhardwaj and V. Bhutani, "Energy Prediction in Urban Areas Using Machine Learning and Deep Learning," 2023 10th IEEE Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON), Gautam Buddha Nagar, India, 2023, pp. 190-195, doi: 10.1109/UPCON59197.2023.10434347.
[8] W. Alomoush, T. A. Khan, M. Nadeem, J. I. Janjua, A. Saeed and A. Athar, "Residential Power Load Prediction in Smart Cities using Machine Learning Approaches," 2022 International Conference on Business Analytics for Technology and Security (ICBATS), Dubai, United Arab Emirates, 2022, pp. 1-8, doi: 10.1109/ICBATS54253.2022.9759024.
[9] M. I. A, N. G, P. Karthikeyan and M. Venkateshkumar, "Forecasting of City Power Consumption for Efficient Energy Management – Smart City," 2025 11th International Conference on Electrical Energy Systems (ICEES), Chennai, India, 2025, pp. 391-396, doi: 10.1109/ICEES67011.2025.11213429.
[10] S. S. Helli, S. Tanberk and O. Demir, "Forecasting Energy Consumption Using Deep Learning in Smart Cities," 2022 International Conference on Artificial Intelligence of Things (ICAIoT), Istanbul, Turkey, 2022, pp. 1-6, doi: 10.1109/ICAIoT57170.2022.10121846.
[11] R. Panigrahi, N. R. Patne, S. Pemmada and A. D. Manchalwar, "Prediction of Electric Energy Consumption for Demand Response using Deep Learning," 2022 International Conference on Intelligent Controller and Computing for Smart Power (ICICCSP), Hyderabad, India, 2022, pp. 1-6, doi: 10.1109/ICICCSP53532.2022.9862353.
[12] S. H. Lee, T. Lee, S. Kim and S. Park, "Energy Consumption Prediction System Based on Deep Learning with Edge Computing," 2019 IEEE 2nd International Conference on Electronics Technology (ICET), Chengdu, China, 2019, pp. 473-477, doi: 10.1109/ELTECH.2019.8839589.
[13] P. Adhikari, F. Prakash and B. Sharma, "A Brief Review on Accuracy Level of Smart and Micro Grid Systems Use of ML and DL Algorithms Through Hybrid Integration," 2023 International Conference on Power Energy, Environment & Intelligent Control (PEEIC), Greater Noida, India, 2023, pp. 528-532, doi: 10.1109/PEEIC59336.2023.10451447.
[14] Z. Wang, S. Wang, Z. Zhou, S. Wei, G. Zhu and F. Wen, "Advanced Renewable Energy Power Prediction and Optimization Configuration Technology of Energy Storage System Capacity," 2023 International Conference on Power System Technology (PowerCon), Jinan, China, 2023, pp. 1-6, doi: 10.1109/PowerCon58120.2023.10331277.
[15] E. Campagne, I. Zehavi, Y. Amara-Ouali, Y. Goude and A. Kalogeratos, "Graph Neural Networks for Electricity Load Forecasting," 2025, doi: 10.48550/arXiv.2507.03690.